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Assessing the accuracy of probabilistic population forecasts

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ABSTRACT

Stochastic demographic forecasts indicate to their users the ex ante level of uncertainty to be expected. There are approximately 25 years' worth of data for population stocks and flows on the performance of such forecasts. The demographic context has numerous special features. The data come in the form of counts. The vital processes of births and deaths, and migration, combine to produce the future counts, so it is not obvious how their separate effects might be assessed. Finally, the ex post assessment is hampered by quality problems in the official population data. These issues are tackled in the context of a stochastic forecast of Norway for the years 2003–2023. A novel approach infers predictions of births, deaths, and net migration from changes in predicted population stocks in carefully chosen age segments. We use the deviance as a scoring rule to measure the lack of fit between the predictive distribution and the realized value. For a 20-year forecast horizon, predictive distributions for deaths in ages 80 years and over are less accurate than the birth distributions. The situation is the reverse for ages below 75 years. The accuracy of the predictive distribution for births appears to deteriorate much more rapidly than that of age-specific deaths for forecast horizons beyond 20 years. Forecasts for net migration are systematically less accurate than those for births or deaths.

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1. Introduction

Predictions of future numbers of births and deaths, and of the size of migration flows, are highly desirable in policymaking. If mortality predictions turn out to be too high, then allocations for old-age pensions will be too low, and addressing the ensuing shortfalls may be politically difficult. When future immigration flows are larger than predicted, there will be added demand for housing, but also an increased supply of labor. Forecasts of births are initially needed for childcare and kindergartens at the municipal level, but over the medium and long term, the effects on the national labor supply dominate. Uncertainties in forecasting have a direct impact on the tax rate that is necessary for balancing a country's public budget in the future (Alho & Lassila, 2023).

Several studies have evaluated the ex post accuracy of past demographic forecasts against observed data. For OECD countries, the studies have shown ostensibly smaller errors for mortality than for fertility, while migration predictions have been the least accurate (Keilman, 1997; NRC, 2000; Wilson, 2012). The possibility of errors was not entirely unexpected early on Whelpton et al. (cf. 1947), but it has been difficult for the public, experts, and even the demographic community to accept that the errors in current forecasts could be as large as those of the past.

In order to provide the users of demographic forecasts with an indication of the ex ante uncertainty of forecasts, we, among others (Alho & Spencer, 1997; Bijak et al.,

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2015; Keilman, 1997), have called for the production of stochastic population forecasts that indicate how much uncertainty one can expect in future demographic developments. In practice, both the point forecast and its uncertainty can be given in terms of prediction intervals, for example.

Finding statistical models for the vital processes that predict intervals of realistic width has been a challenge. Mortality has generally declined, but at varying rates and with shocks (such as Covid-19) interspersed. Migration has gone through alternating periods of high and calm levels relative to political and economic developments in the sending countries, but with a sustained growing trend. Fertility has been clearly non-stationary, with characteristics that are not easily replicated by parsimonious time-series models. Most recently, there has been an unexpected decline in northern Europe, the U.S., South Korea, and China.[1] Yet, historical data and common sense suggest that there are homeostatic mechanisms that keep the vital processes in check. Ad hoc attempts to keep, e.g., the predicted fertility variables within bounds have included logit transformations, rejection sampling, and absorbing boundaries, for example (Alho, 1990; Keilman, 2002; Keilman et al., 2002).

We take up the question of the relative accuracy in the context of a more general discussion of the accuracy of probabilistic demographic forecasts.[2] Fertility, mortality, and migration are all *counts* which separates them from other processes that are also forecasted using probabilistic methods, such as weather or sea level. Among themselves, the three processes are also quite different components of population change. The births of a calendar year affect only the youngest age group next year (newborns), whereas deaths and migration impact all ages, one by one, including earlier surviving births. The fact that the effects of the three vital processes become superimposed is a genuine logical problem for any attempt to disentangle their separate effects (e.g. Shryock et al., 1976).

However, by restricting the analysis of fertility to the newborns only, the effects of mortality and migration are greatly reduced. Although migration does occur in all ages, it is primarily concentrated in young-adult ages (say, 20–35 years old). In these ages, mortality is quite low, so a serviceable analysis of the accuracy of migration forecasts can be obtained from annual changes in those ages. Mortality is the dominant source of uncertainty in older ages (say, ages 65+), and the accuracy of mortality forecasts can be obtained from annual changes in those ages, because migration is very low. We believe that the non-trivial approach of handling demographic bookkeeping in terms of flows, to be developed more fully below, is a major contribution. In particular, we believe that the handling of net migration is novel.

[1] These stylized facts are intended to characterize OECD countries after the so-called demographic transition, i.e. roughly after World War I.

[2] Such forecasts may equivalently be called stochastic forecasts when the role of modeling is emphasized, or predictive distributions when the emphasis is on the entire future development of the population vector.

The ex post facto evaluations mentioned above apply to deterministic forecasts. The typical measures of error include the signed error, absolute error, relative error, and mean squared error. They are based on a comparison of point forecasts and the realized values. Besides the point forecast, no other aspect of the predictive distribution of the future population is taken into account.

In this paper, we develop a statistical approach for evaluating the accuracy of probabilistic forecasts of births, deaths, and net flows of international migration, when actual counts for these three types of events are available. As far as we know, this is the first study that assesses probabilistic forecasts for these events. There is one earlier ex post evaluation of probabilistic demographic forecasts, namely (Keilman, 2020). Keilman analyzed predictive distributions of total population size and of the age composition, but not of births, deaths, or migrations. He used goodness-of-fit criteria based on the first two moments of the predictive distributions, on the width of prediction intervals, and on simulated complete predictive distributions that were available as a sample.

The current analysis is part of a much broader stream of literature on the verification of probabilistic forecasts in various fields such as meteorology, finance, healthcare, and risk management. Key verification aspects include calibration, which compares predicted probabilities with frequencies, and sharpness, which evaluates the concentration of forecasts. The degree of lack of fit is often expressed by computing a scoring function. Examples are the Brier score (Brier, 1950), which is useful for categorical forecasts; the continuous ranked probability score (CRPS; Keilman, 2020; Matheson & Winkler, 1976), which assesses probabilistic forecasts in the form of cumulated density functions; and the logarithmic score (Good, 1952). Gneiting and Raftery (2007) and Gneiting and Katzfuss (2014) review the field. The verification of probabilistic forecasts plays a vital role in improving forecasting models and ensuring their usefulness for a decision maker.

As in any assessment of this type, our analysis of the accuracy of demographic forecasts implies a tradeoff between the location and scale of the predictive distribution. If the location (here, a point forecast) is far from the true value, a high value of scale (here, the width of the prediction interval) is needed to cover the observed future value, or else the purported ex ante prediction intervals will be unrealistically narrow. In other words, there is a tradeoff between bias and variance. Whereas many different scoring functions have been proposed (see the discussion above), it turns out that simple concepts of likelihood-ratio-based hypothesis testing can be tailored to the case at hand. That is, we propose using deviance as a scoring rule to measure the lack of fit between the predictive distribution and the realized value. This is closely related to the work of Knorr-Held and Rainer (2001) and Gneiting and Raftery (2007). From a theoretical perspective, likelihood-ratio tests are uniformly the most powerful for single-parameter exponential family distributions (Lehmann, 1986), the case we have for births and deaths. In the corresponding generalized linear models, the tests are used in variable selection and

goodness-of-fit testing, and usually implemented via the deviance. On the other hand, the test statistics are equivalent to the χ^2 tests familiar to most applied statisticians from introductory courses. In our case, such evaluations are needed for very high-dimensional arrays of time- and age-specific predictive distributions. The interdependencies in the counts of births, deaths, and migrations will be addressed by conditioning.

The deviances will be based on Poisson distributions for the counts of the vital events. The Poisson is known to approximate well rare counts from binomials, or more general sums of Bernoulli trials. It arises from the theory of arrival processes and contingency tables. It is also closed under addition, a property that is useful in our application. When one thinks of populations as stocks whose sizes are changed by flows, then it is the latter for which the Poisson model makes sense.

We apply the measure to assess the goodness of fit of predictive distributions for births, deaths, and net migration flows from a historical probabilistic population forecast for Norway for the years 2003–2023. Special attention is paid to the quality of the population data that are used to assess the accuracy of forecasting. As with other Nordic countries, Norway is considered to have population data of the highest international quality. Yet the register-based Norwegian population estimates for a given age and year do evolve over time, primarily because of updates of migration data. Thus, in an assessment of the type we envision, the errors of the database must be addressed.

The paper is structured as follows. In Section 2, we outline how numbers of births, deaths, and net migrations can be approximated from segments of a population's age structure. Such approximations are needed when, as is often the case, predictive distributions for population counts by age are available, but not for births, deaths, and migrations. Section 3 sketches the statistical framework for the scoring rule. The rule itself is derived in Section 4. Section 5 discusses problems that arise when our scoring rule is considered for multiple ages or time intervals. The application that assesses the relative goodness of fit of predictive distributions for counts of births, deaths, and net migration using Norwegian data is in Section 6, and conclusions are in Section 7. Supplementary information is in three appendices. Appendix 1 discusses errors in the Norwegian population register, as well as errors caused by the approximations in Section 2. Appendix 2 gives details for the Poisson-based model for immigration counts. The relationship between our scoring rule and the non-central χ^2 distribution is explained in Appendix 3.

2. Cohort-component bookkeeping

Population counts are the result of the operation of vital processes of birth and death, and of migration (e.g. Shryock et al., 1976). They operate in continuous time, but a discretized version suffices for our purposes. Deaths and out-migration deplete the population. Births and in-migration increase it. These simple bookkeeping principles are formalized in the so-called cohort-component model, which is the standard approach for computing population forecasts (Shryock et al., 1976). The main idea is to update a table with known numbers for the population pyramid, taken from a recent census or from a population register, to a new table one year later. The update requires assumptions on mortality (the shares of men and women of a given age who survive one year later), fertility (the mean number of children per woman born during the year), and migration (e.g. age- and sex-specific numbers of net migration). The parameters for these three processes interact, for instance when computing the number of newborn children who survive to the end of the year or the survival of persons who enter the country. Based on assumptions of this kind for many years in the future, the process can be repeated, resulting in a population forecast that could cover several decades.

This shows that fertility, mortality, and migration combine in complex ways to produce population stocks. However, the flows of births, deaths, and net migration operate almost in separation at certain ages.

In Fig. 1, a Lexis diagram is used to display the relative separateness of the vital processes. Although mortality is present in all ages, in present-day European countries, its effect is dominant in older ages: a lower limit of 55 years is used in the figure, for illustration. Migration is quite low in these ages. Therefore, changes in population stocks and the accuracy of forecasting are almost entirely determined by mortality. In the very highest ages, unexplainable binomial/Poisson variation begins to play an increasingly large role, so in the top polygon of the figure, an upper limit of 90 years is used to delineate a possible region in which the accuracy of mortality forecasting might be studied.

Migration is primarily concentrated in young adults (20–35 years old, in Fig. 1). Mortality is quite low in these ages, so changes in population stocks and the accuracy of forecasting are primarily determined by net migration. The corners of the otherwise rectangular region have been cut to indicate that an aggregation of migration effects might best be done along cohort lines in the middle parallelogram.

Births are a product of the size of the childbearing (female) population and the average level of fertility in these ages. Thus, when one controls for the size of the childbearing population, births are essentially determined by fertility.[3] While mortality and migration certainly operate during the first year of life, particularly during the first half of the previous century when infant mortality was high, the major uncertainty nowadays pertains to fertility.

To formalize these observations, we define $V(x, t)$ as having population counts by age $x = 0, \ldots, \omega$ in the beginning of year $t = \cdots, -1, 0, 1, 2, \ldots$. Cohort-component bookkeeping means that

$$V(x+1, t+1) = V(x, t) - D(x, t) + I(x, t) - O(x, t), \quad (1)$$

[3] The cohort-component approach distinguishes the population according to age and sex only. When there is a considerable share of foreign-born women of childbearing age, and these women have much higher fertility than native women, births are determined by migration flows as well. However, due to the relatively low share of foreign-born women in OECD countries, migration does not greatly affect the level of overall fertility, even in the countries where immigrant fertility is significantly higher than that of natives (Bagavos, 2019).

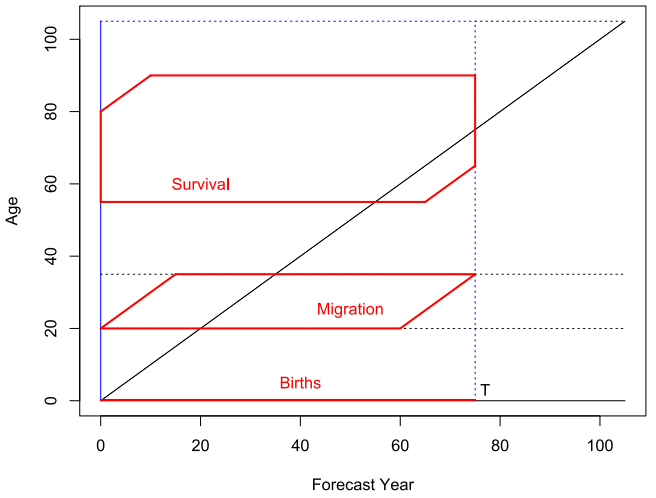


Fig. 1. Areas in a Lexis diagram relevant for the study of forecast accuracy.

where $D(x,t)$ denotes the number of deaths of those who were age x at t, $I(x,t)$ is number of in-migrants of age $x+1$ at $t+1$, and $O(x,t)$ is the number of out-migrants to all other countries of age x at t. In old age, net migration $N(x,t) = I(x,t) - O(x,t)$ is very small, so for older adults, we have

$$V(x+1, t+1) \approx V(x,t) - D(x,t). \tag{2}$$

In contrast, for young adults, $D(x,t)$ is very small, so we have that

$$V(x+1, t+1) \approx V(x,t) + N(x,t). \tag{3}$$

A separate consideration of in- and out-migration can be illuminating, but detailed data on gross migration flows are typically of poor quality, or lacking. More importantly, their net effect is in most OECD countries positive, or the total net migration $N(t) \equiv \sum_x N(x,t) \gg 0$.

The formulas above hold, as such, separately for males and females. For the purpose of describing births, assume that the population of age x, below, refers to females only. Mortality and net migration during the year of birth are small, so the births can then be written as

$$V(0, t+1) \approx \sum_x B(x,t), \tag{4}$$

where $B(x,t)$ denotes the number of births born to the $V(x,t)$ women who were age x at t. Childbearing age is often taken to be $x = 15, \ldots, 49$, for example.

In Appendix 1, we use data for the population of Norway for ages x = 0, 1, 2, …, 100 and years t = 2003, 2004, 2005, …, 2022 to assess the quality of approximations (1)–(4). We show that death counts are well approximated by the change in population stocks for age 60 years and over. On average across the years, the error for these ages is limited to between −4 and +4 per cent compared to actual death counts. Errors in net migration counts are less than 5% for ages 20–30 years, but they increase rapidly to values of 10% or more for higher ages. Approximating births by the population aged 0 at the end of the year implies errors of less than 1%. These findings relate to empirical data. It is reasonable to assume that they also hold for the outcomes of the probabilistic forecast in Section 6.

3. Poisson approximation for demographic flows

3.1. Deaths

Consider first a chain of binomial experiments, indexed by time in older adults. In a closed cohort, (2) would hold exactly. Suppose it does. Assume that conditionally on $V(x,t)$, the deaths have a binomial distribution $D(x,t) \sim Bin(V(x,t), \pi(x,t))$. Then, conditionally on $V(x+1, t+1) = V(x,t) - D(x,t)$, assume that $D(x+1, t+1) \sim Bin(V(x+1, t+1), \pi(x+1, t+1))$. Thus, the latter binomial model depends on the outcome of the first.

Since the populations $V(x,t)$ are typically large (in the thousands, say), and the probabilities of death are small (say, 0.001–0.1), the binomial distributions are well approximated by the Poisson distribution (Feller, 1968), or $D(x,t) \sim Po(\lambda(x,t))$,[4] with expectation $\lambda(x,t) = V(x,t)\pi(x,t)$. An important simplification that follows is that the expectation and the variance of the distribution are equal.

3.2. Net migration

Following the principles laid out above, the out-migrants have an approximate Poisson distribution $O(x,t) \sim Po(V(x,t)o(x,t))$, where $o(x,t)$ is the out-migration probability. In-migrants are a sum of out-migrants from all other countries. Each of those has a similar Poisson distribution. Instead of introducing region-specific notation throughout the main text, a brief argument is provided in Appendix 2. The result is that if the migration flows are independent, with expectations proportional to the size of the receiving populations, we have the approximate result

[4] $Y \sim Po(\lambda)$, if the probability that $Y = y$ equals $e^{-\lambda}\lambda^y/y!$, for $y = 0, 1, 2, \ldots$

$I(t) \sim Po(V(x,t)\iota(x,t))$, where $\iota(x,t)$ is a sum of country-specific out-migration intensities towards our country of interest.

The net migration $N(x,t) = I(x,t) - O(x,t)$ is seen to be a difference of two Poisson-distributed variables. Assuming that they are independent, $N(x,t)$ has an approximate normal distribution, with expectation $V(x,t)(\iota(x,t) - o(x,t))$ and variance $V(x,t)(\iota(x,t) + o(x,t))$. The expectation is typically positive and observable, but the variance depends on gross migration and may not be available.

When detailed information about in- and out-migration is not available, and $\iota(x,t) \gg o(x,t)$, a shortcut approach is to assume that $N(x,t) \sim Po(V(x,t)\nu(x,t))$, where $\nu(x,t) = \iota(x,t) - o(x,t)$ is the net migration rate. This has the virtue of unifying the assessment of the accuracy of forecasting.

3.3. Births

The random count of births $B(x,t)$ differs slightly from the above cases, because of the possibility of multiple births (twins, triplets, etc.) or repeats during the same calendar year. This suggests that a simple binomial model is not the most realistic starting point. However, viewing births as being produced by an arrival process, a direct Poisson assumption is expedient, $B(x,t) \sim Po(V(x,t)\phi(x,t))$ conditionally on $V(x,t)$, where $\phi(x,t)$ is the age-specific fertility rate (Alho & Spencer, 2005).

Starting from the Poisson assumption for each age x, and assuming them to be independent, we get that $V(0,t) \sim Po(V^{cb}(t)\phi(t))$, where $V^{cb}(t) = \sum_x V(x,t)$ is the population in childbearing age. The total births are $B(t) = \sum_x B(x,t)$ with expectation $E[B(t)] = \sum_x V(x,t)\phi(x,t)$, and $\phi(t) = E[B(t)]/V^{cb}(t)$ is the average fertility rate, also known as the general fertility rate (Shryock et al., 1976).

4. Likelihood-ratio-based goodness of fit

4.1. Nature, forecaster, and assessor

We consider a game in which Nature generates the population counts. The Forecaster attempts to guess the counts just after time $t = 0$, using a predictive distribution. Past population history for $t = \cdots, -1, 0$, is available for him. The Assessor then measures different aspects of the accuracy of the forecasts of flows, or any functionals of the flows.

A constraint Nature has is that the population counts must obey the cohort-component bookkeeping (1). She also must use the chain binomial models to generate the flows (or Poisson for the births), with the parameters $\pi(x,t)$, $\iota(x,t)$, $o(x,t)$, $\phi(x,t)$, of her own choosing, from the jump-off counts $V(x,0)$ that are also known to the forecaster.[5]

[5] This assumes that errors in the jump-off counts of the forecast can be ignored. This is a reasonable assumption in our empirical application with Norwegian data. However, historical population forecasts of many other countries show jump-off errors of appreciable sizes (e.g. NRC, 2000) .

The eventual aim is to compare alternative forecasts of the same population development by different forecasters, or forecasts of different population developments. This could refer to forecasts of one country at different times, forecasts of different countries at a given time, or, as in our application in Section 6, forecasts of different vital processes at any given time. The feasibility of such a wide range of applications depends, in a major way, on the Poisson assumption we make.

The Forecaster may use past demographic data for $t = \cdots, -2, -1$ and population counts $V(x,0)$ in whatever way he pleases to forecast the parameters used by Nature, and thereby the expected future population flows and stocks. The difference between the binomial/Poisson expectations and the actual counts is not forecastable, by assumption. This point of view agrees with the practices of demography and actuarial science, where the best one can hope for is to get, say, the expected counts of deaths right. Unexplainable random variation always remains because the identities of those who actually die cannot be known in advance.

The Forecaster typically uses statistical time-series modeling and judgment, in combination. To allow the forecast users to appreciate the uncertainty of the forecast, the Forecaster produces a predictive distribution to express the forecast. The practicalities are, however, such that usable predictive distributions can be expressed in analytical form only for a limited subset of population flows. Instead, a stochastic simulation is used to obtain a numerical approximation to the underlying probability distribution. Thus, the Forecaster produces samples $V_{(i)}(x,t), i = 1, \ldots, N$ by stochastic simulation from his assumed predictive distribution.

It is assumed that as a minimum, the Assessor has available the full set of sizes of population stocks in all ages for both sexes, and for all forecast years for each sample. Alternatively, the Assessor may have direct access to population flows. The Assessor is disinterested in the sense that some types of errors may be analyzed today, something else tomorrow. Consistency across assessments is required.

The task is to measure the accuracy of the Forecaster's predictive distribution based on the simulations, after the fact, when the future development has unfolded. The predictive distribution may be right on target, on average, but have far too wide a spread, or it may have a small spread but not cover the target, for example. Both bias and error variance matter. This is where the scoring rules (here, deviances) come into play.

4.2. Deaths

As discussed above, mortality is the dominant determinant of change in population size from one age to the next in the late-adult ages. For such ages x, we assume as a bare minimum to have stock data available, so from the approximate identity $V(x,t) - V(x+1,t+1) \approx D(x,t)$, an estimate of actual death counts can be derived. Alternatively, we may have a direct estimate of $D(x,t)$ available. In the case of Norway, direct register-based evidence about annual deaths is collected and believed

to be highly accurate; cf. Appendix 1. In either case, an estimate of the count $D(x,t)$ is assumed to become known to the Assessor when the future has unfolded.

Using the Poisson approximation to the distribution of $D(x,t)$, the true log likelihood of the observed value is

$$\ell(\pi(x,t)) = -V(x,t)\pi(x,t) + D(x,t)\log(V(x,t)\pi(x,t)) \quad (5)$$

up to an additive constant. (In case, $V(x,t) = 0$, also $D(x,t) = 0$, and the Poisson distribution $Po(0)$ is defined as putting a unit mass on 0. In this case, (5) equals 0.) This is known only to Nature. However, (5) is maximized at $\pi(x,t) = d(x,t)$, where $d(x,t) = D(x,t)/V(x,t)$. This is the maximum likelihood estimator. Since we are considering a single observation from a single Poisson trial, this is also the maximum likelihood estimator under the so-called saturated model that has one parameter per observation, by definition. The maximum value is

$$\ell(d(x,t)) = -V(x,t)d(x,t) + D(x,t)\log(V(x,t)d(x,t)). \quad (6)$$

Should it happen that $V(x,t) > 0$, and $D(x,t) = 0$ so $d(x,t) = 0$, then (6) would equal $-V(x,t) \times 0 + 0 \times \log(V(x,t) \times 0) = 0 + 0 \times \log(0) = \log(0^0) = \log(1) = 0$.

For each sample i, the Assessor may use stock data and compute the forecasted probability of death as $d_{(i)}(x,t) = (V_{(i)}(x,t) - V_{(i)}(x+1,t+1))/V_{(i)}(x,t)$.[6]

If $d_{(i)}(x,t)$ were the true value of $\pi(x,t)$, the log likelihood would be

$$\ell(d_{(i)}(x,t)) = -V(x,t)d_{(i)}(x,t) + D(x,t)\log(V(x,t)d_{(i)}(x,t)). \quad (7)$$

For precision, note that apart from rounding the simulation results, we always have $d_{(i)}(x,t) > 0$ under the model PEP uses. As noted above, the usual log-likelihood ratio statistic for sample i is then

$$R_{(i)}(x,t) = 2(\ell(d(x,t)) - \ell(d_{(i)}(x,t))). \quad (8)$$

In other words, this is the test statistic one would use to test the null hypothesis $H_0 : \pi(x,t) = d_{(i)}(x,t)$ against the alternative hypothesis $H_A : \pi(x,t) \neq d_{(i)}(x,t)$. It is well known that if the null hypothesis were true, the statistic would have a χ^2 distribution with one degree of freedom. However, we use the statistic as a measure of goodness of fit for a range of samples i that obviously cannot be simultaneously correct. Thus, for each sample i, one may think of there being a non-central χ^2 distribution (e.g. Johnson et al., 1995). The variation of the non-centrality parameters is a major component of the lack of fit. In Appendix 3, we derive and discuss the non-central χ^2 distribution.

Averaging the statistics over samples provides the desired lack-of-fit statistic:

$$\bar{R}(x,t) = \frac{1}{N}\sum_{i=1}^{N} R_{(i)}(x,t). \quad (9)$$

[6] An alternative estimate may be available, if the simulations can be run by setting net migration to zero. In the case of PEP (the program that was used to compute the probabilistic forecast in Section 6), this is doable but requires detailed knowledge of how the program has been parameterized. An intermediate third possibility is to use the actually observed net migration values to yield a migration-adjusted estimate $d_{(i)}(x,t) = (V_{(i)}(x,t) - V_{(i)}(x+1,t+1) - N(x,t))/V_{(i)}(x,t)$.

In words, $\bar{R}(x,t)$ equals two times the logarithm of the geometric mean of the likelihood-ratio statistics for the hypotheses $H_0 : \pi(x,t) = d_{(i)}(x,t), i = 1,\ldots,N$. Large values indicate a large lack of fit.

Suppose there are two different forecasts of a single mortality probability $\pi(x,t)$ based on the values $V(x,t)$ and $D(x,t)$, resulting in respective goodness-of-fit measures $\bar{R}^{(1)}(x,t)$ and $\bar{R}^{(2)}(x,t)$. The smaller value would designate the winner, because it would indicate who was closer to the best possible saturated model.

For precision, we write (8) as $R_{(i)}(x,t) = 2([\ell(d(x,t)) - \ell(\pi(x,t))] + [\ell(\pi(x,t)) - \ell(d_{(i)}(x,t))])$. The first term in square brackets is always non-negative, so it has a non-negative expected value under any distribution. The second term has a non-negative expectation under the model $Po(V(x,t)\pi(x,t))$ that Nature is assumed to use to generate the deaths. The Kullback–Leibler divergence between the true distribution of deaths $Po(V(x,t)\pi(x,t))$ and any other discrete distribution with the same support, say G, is non-negative and vanishes only when $G = Po(V(x,t)\pi(x,t))$. If the Assessor had access to the predictive distribution Nature uses, he could use the latter score to assess lack of fit whenever $V(x,t) > 0$; if $V(x,t) = 0$, then $D(x,t) = 0$ with probability one.

By the Neyman–Pearson lemma, likelihood-ratio tests are uniformly the most powerful unbiased tests for exponential families (see Thm. 3, hypothesis H4 of Lehmann, 1986, p. 147). As the Poisson is a member of this class of distributions, the result supports the construction we propose for assessing lack of fit. This optimality is further detailed by Feuerverger and Rahman (1992), who consider the assessment of forecasts of binary outcomes by different forecasters.

Apart from rounding, simulated births and deaths are strictly positive. In Section 6.2 we explain the way we handled zero and negative values for migration counts.

Example 1. Consider deaths in a population of size $V(x,t) = 1,000$. Suppose the count turned out to be $D(x,t) = 43$. This means that the (Poisson approximation to the) log likelihood is $\ell(d(x,t)) = -1000 \times d(x,t) + 43 \times \log(1000 \times d(x,t))$ (up to a constant). Prior to observing the deaths, five forecasts were made in terms of beta distributions. Fig. 4 displays their densities. The blue curve is the density for $Be(4,96)$. It has an expected value of 0.040 and a standard deviation of 0.0195. The other, red densities correspond to four values of a parameter that influences the spread of the distribution: $n(1) = 0.5$, $n(2) = 1$, $n(3) = 10$, $n(4) = 100$. All of these have the mean 0.045, but the standard deviations decrease as a function of $n(i)$. They are 0.0290, 0.0206, 0.0093, and 0.0041.

The average of log likelihoods (7) for all the five forecasts can now be computed. In other words, the averages are obtained by integrating over $(0,1)$, the product of the log likelihood in Fig. 3 and each of the densities in Fig. 2. The average for $Be(4,96)$ is 113.25. For the other four, they are 108.85, 113.96, 117.77, and 118.50. The finding is that although the four latter forecasts are, on average, closer to the actual result, which is given by the black solid vertical line at 0.043, the one with the largest standard deviation (and variance) has a lower average log

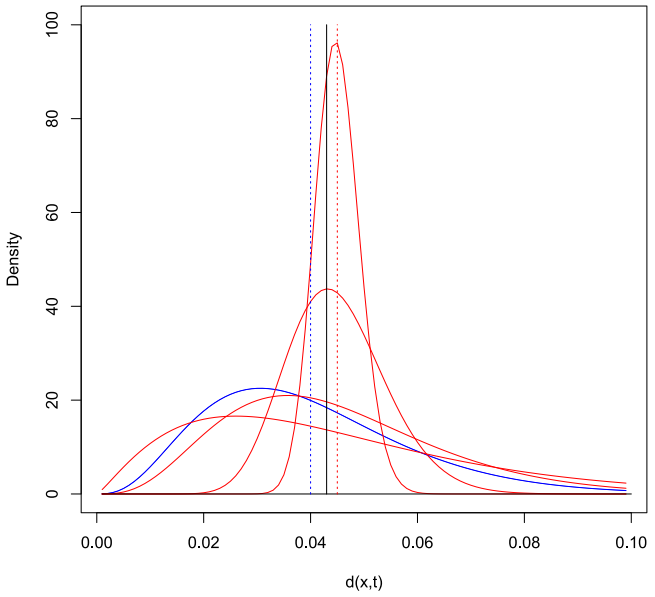


Fig. 2. Densities of stochastic forecasts: $Be(4, 96)$ (blue), and $Be(n(i) \times 4.5, n(i) \times 95.5), i = 1, 2, 3, 4$ (red). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

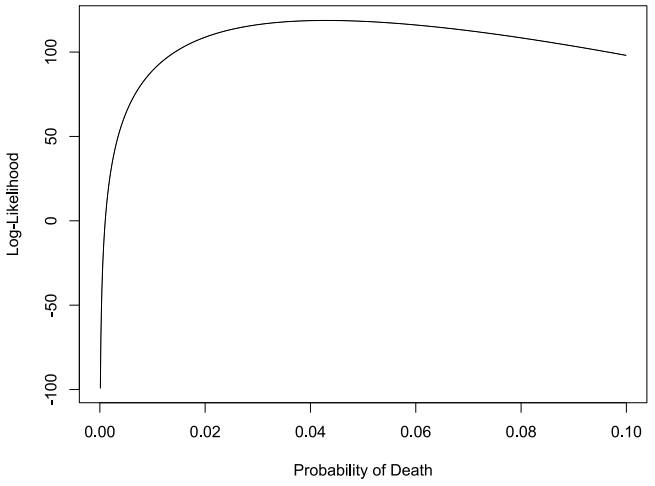


Fig. 3. Log likelihood of the data.

likelihood (and hence a larger score $\bar{R}(x, t)$) than the blue density. This is an example of the bias–variance tradeoff that is the central aspect in forecast evaluation.

A reader wishing an exact connection of this example, and Eqs. (7)–(9), may think of the here analytically given beta distributions as toy examples that correspond to the analytically intractable distribution the Forecaster uses. When the Forecaster takes N samples of his predictive distribution, this is the same thing as when we take N samples from the beta distributions and have Eqs. (7)–(9). Indeed, the numerical values displayed in Fig. 2 were computed in this way—as sample averages. That is, stochastic integration was used with $N = 1,000,000$. Of course, in practical examples, much smaller values are currently used in stochastic forecasting, such as $N = 3,000$.

4.3. Net migration

When we make the Poisson distribution assumption $N(x, t) \sim Po(V(x, t)$
$\nu(x, t))$ for net migration, we are back in the case already handled for deaths, only now $N(x, t)$ takes the role of $D(x, t)$, $\nu(x, t)$ takes the role of $\pi(x, t)$, and the age range for x comprises young adults (say, in the middle parallelogram of Fig. 3). The fundamental difference is that in this case, $N(x, t) = V(x + 1, t + 1) - V(x, t) \gg 0$, because mortality is close to zero and in-migration is assumed

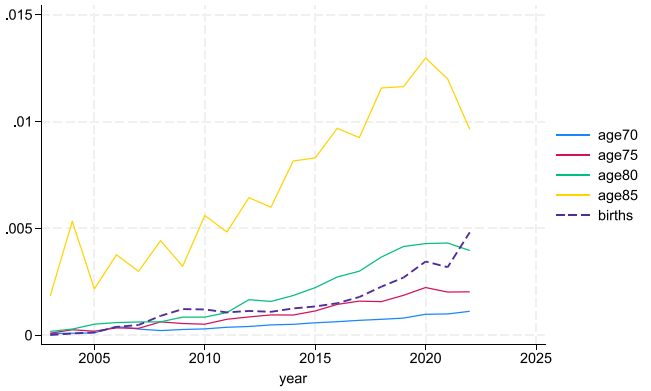


Fig. 4. Goodness-of-fit statistics $\tilde{R}$ for predictive distributions of deaths at ages 70, 75, 80, and 85 years (solid colored lines), and births (dashed dark blue line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to be much higher than out-migration. As in the case of mortality, a more accurate direct estimate of net migration may be available. In either case, under the saturated model, the migration rate is $n(x,t) = N(x,t)/V(x,t)$ and the corresponding log likelihood is

$$\ell(n(x,t)) = -V(x,t)n(x,t) + N(x,t)\log(V(x,t)n(x,t)). \tag{10}$$

The estimate of migration rate for sample i is $n_{(i)}(x,t) = (V_{(i)}(x+1,t+1) - V_{(i)}(x,t))/V_{(i)}(x,t)$.[7]

The corresponding log likelihood is

$$\ell(n_{(i)}(x,t)) = -V(x,t)n_{(i)}(x,t) + N(x,t)\log(V(x,t)n_{(i)}(x,t)). \tag{11}$$

The computation of the average log likelihood goodness-of-fit statistic follows, as above.

4.4. Births

The Poisson assumption for births is $V(0,t+1) \sim Po(V^{cb}(t)\phi(t))$. In the saturated model, the average birth rate is $b(t) = V(0,t+1)/V^{cb}(t)$.[8] Using the adopted notation, this would mean taking $b(t) = B(t)/V^{cb}(t)$ as the observed birth rate. In either case, the log likelihood is

$$\ell(b(t)) = -V^{cb}(t)b(t) + V(0,t+1)\log(V^{cb}(t)b(t)). \tag{12}$$

In this case, the average fertility in simulation i is $b_{(i)}(t) = V_{(i)}(0,t+1)/V^{cb}_{(i)}(t)$. Two alternatives may be available, as in the case of mortality. The corresponding log likelihood is

$$\ell(b_{(i)}(t)) = -V^{cb}(t)b_{(i)}(t) + V(0,t+1)\log(V^{cb}(t)b_{(i)}(t)). \tag{13}$$

The rest follows verbatim.

[7] The small bias caused by mortality could be removed using a death-adjusted estimate $n_{(i)}(x,t) = (V_{(i)}(x+1,t+1) - V_{(i)}(x,t) + D(x,t))/V_{(i)}(x,t)$.

[8] As in the case of deaths, we may have a direct estimate of births.

5. Notes on aggregation

5.1. Combining cells

As noted in Section 3.1, when followed over cohort lines, the counts of deaths are produced as a chain of binomial trials. Results from such a chain can be simply added, without introducing additional assumptions. To see this, let us start from some age x and time t (within the top polygon in Fig. 3) that has population $V(x,t)$. Assume that this count is known. We define events $A(x+k,t+k) = \{D(x+k,t+k) = D_{x+k,t+k}\}$, where $D_{x+k,t+k}$ denotes some counts of deaths. Suppose we follow the cohort $K+1$ steps. By repeated conditioning, we obtain the result

$$P(A(x,t), A(x+1,t+1), \ldots, A(x+K,t+K) \mid V(x,t)) = \prod_{k=0}^{K} P(A(x+k,t+k) \mid V(x+k,t+k)), \tag{14}$$

because, e.g., $V(x,t)$ and $D(x,t)$ determine $V(x+1,t+1)$; this and $D(x+1,t+1)$ determine $V(x+2,t+2)$, etc. The probabilities $P(A(x+k,t+k) \mid V(x+k,t+k))$ are precisely the binomial probabilities we started out with. Taking the logarithm of (14), the product is transformed into a sum. Thus, the distance measures (9) corresponding to each of the $K+1$ terms are simply added to obtain a measure for the error in the forecast of cohort deaths during the $K+1$ years.

The chains of binomial trials for different cohort lines are independent by our assumptions, so their log likelihoods can be added at will.

The corresponding argument for x and t in the middle parallelogram of Fig. 1 has exactly the same structure under the assumptions we have made. The argument will not be repeated.

Aggregating births (or the number of those in age 0) during the first 15 (or so) forecast years presents no similar need to consider repeated conditioning. The counts are independent, given $V^{cb}(t)$. After that, earlier births gradually begin to influence $V^{cb}(t)$. This effect can be accounted for by repeated conditioning, and we are back

to the case already handled. The result is that the distance measures (9) can simply be added for the birth years of interest.

5.2. Weighting

When measures (9) are added over selected values of x and t (to cover, e.g., the top red polygon of Fig. 1) one may worry about the comparability of the goodness-of-fit measures in different ages, and over time. Indeed, the value of (8) changes if one multiplies the values of $V(x,t)$ and $D(x,t)$ by some positive constant $c \neq 1$. This makes it difficult to compare the goodness of fit of mortality between, for example, young adults and the elderly of around 80 years old, where the numbers of deaths may differ by one order of magnitude. Migration flows differ strongly by age as well, with very low levels for ages around 15 years, and much larger flows for ages 20–35. Business cycles and political upheaval cause migration to fluctuate strongly over time.

Here, the Assessor may have different points of view. When the overall interest is in population counts that are determined by the flows, i.e. if all cells (x,t) are desired to have the same, or at least similar, weight in the summary measure, this can be achieved by fixing $D(x,t) \equiv 1$ and adjusting the values of $V(x,t)$ accordingly. In that case, the fit of the predictive distributions for mortality or net migration will be comparable between ages. Presumably, other weightings can be entertained,[9] but choosing transformed variables $\tilde{D}(x,t) \equiv 1$, and then $\tilde{V}(x,t) = V(x,t)/D(x,t)$ has the consequence: that the log likelihood of the saturated model (6) becomes -1 for all (x,t) under consideration. In other words, the maximum value of the log likelihood (including the constant that was omitted from the formulas) -1 is the same for all cells. Yet the maximum likelihood estimators remain unchanged: $\tilde{d}(x,t) = \tilde{D}(x,t)/\tilde{V}(x,t) = d(x,t)$.

Alternatively, if the Assessor wishes to emphasize the accuracy of the forecasts of the vital rates, then the forecasts of these can be assessed by dividing (8) by $V(x,t)$. In this case, the role of $V(x,t)$ is entirely eliminated, because the terms $\log(V(x,t))$ cancel. The result for the scaled log-likelihood ratio statistic for sample i is then

$$\tilde{R}_{(i)}(x,t) = 2(d_{(i)}(x,t) - d(x,t) - d(x,t)\log(d_{(i)}(x,t)/d(x,t))), \tag{15}$$

and likewise for fertility and migration. The same result is obtained if one fixes $V(x,t) \equiv 1$.

6. Application: Is mortality easier to predict than fertility or migration?

6.1. Uncertain population of Europe

We used a probabilistic forecast for the population of Norway to illustrate these ideas. In particular, we evaluated predictions of births, deaths, and net migration. A view that mortality is easier to predict than fertility or migration has been expressed in various ways for decades (e.g. Keilman, 1997, p. 260, O'Neill et al., 2001, p. 266, Vallin & Caselli, 2004, p. 346; see also Myers, 1955, p. 110, and UN, 1955, p. 296, for early references).

The forecast that we evaluate was computed in 2004 as part of the project Uncertain Population of Europe (UPE). This European Union-funded project[10] was carried out by researchers from Finland, the Netherlands, and Norway; see Alders et al. (2005). Its purpose was to calculate probabilistic population forecasts for 18 European countries, including Norway. Uncertainty was quantified such that it reflected historical variations in fertility, mortality, and international migration. The method used was the model for scaled forecast errors (the scaled model of error; see Alho, 1998 and Alho & Spencer, 2005), with the associated simulation PEP (the Program for Error Propagation)—a menu-driven program that simulates the scaled model of error. Starting in 2003, it was used to compute predictive distributions for numbers of men and women by age (0, 1, 2, …, 100+ years old) for each of the 18 countries, and each year in the period from 2004–2050.

The model for scaled forecast errors assumes that the vital rates are normal in the log scale, and that net migration numbers are normal in the original scale. The normal distribution requires that one specify the mean of the distribution as a measure of location, and the standard deviation as a measure of spread ('scale') around the mean, to reflect uncertainty. Correlations across ages and across time, as well as between men and women, have to be specified. Fertility, mortality, and migration are assumed to be independent.

In the UPE project, the situation in 18 European countries was analyzed concurrently in order to obtain robust parameter estimates. The estimates were based on three sources of information: first and foremost, statistical analyses of observed trends; second, empirical errors in historical forecasts of 14 countries; and third (for international migration in particular), partly also on expert opinions.

For mortality, it was assumed that eventually for each country, the rate of decline in age- and sex-specific mortality rates would converge towards a European average rate of decline, to occur by the year 2030. The eventual rate of decline was empirically estimated from data during the past 30 years for 11 countries with mortality statistics of good quality. The initial rate of decline started from the observed decline for each country. This gave point predictions for death rates in each country for men and women by single years of age. The standard deviation of the forecast error was based on empirical errors in naive forecasts for nine European countries.

As to fertility, the UPE researchers noted a clear geographic division in fertility levels in Europe. The UPE countries in northern and western Europe had experienced levels of completed fertility and total fertility of about 1.8 children per woman. Mediterranean and German-speaking countries were at historically low levels

[9] This is where the discussion of the non-central χ^2 distribution becomes relevant. See Appendix 3 for details.

[10] Grant nr. SERD-2000-00172.

of around 1.4 children per woman. For the period from 2003–2050, it was assumed that these clusters would remain essentially at their current levels. Based on empirical estimates from six countries with long data series, the forecasters found that the logarithm of total fertility behaved roughly like a random walk process.

Migration was expressed as net migration per thousand of the population of 2000. A linear trend model showed a significant upward linear trend in net migration for many countries. However, a mechanical extrapolation would result in a level of net migration for the whole of the 18 countries equal to 5 per thousand in 2050. This was deemed unrealistically high. As a compromise, a value of 3.5 per thousand was adopted, still considerably higher than the aggregate level for the 18 countries in 2003, which was 2.7 per thousand. Next, long-term country-specific assumptions on net migration were made that varied from 1.5 (Finland, France, Iceland) to 6 per thousand (Luxembourg). For Norway, the assumed level in 2050 was 3.5 per thousand.

Details about the mean values of the distributions of fertility and mortality rates and of net migration are provided by Alders et al. (2007) and Keilman et al. (2008). Alho et al. (2008) describe the scaled model in more detail and explain how the standard deviations around the mean were specified using empirical data.

6.2. Goodness of fit of fertility, mortality, and migration

Our interest is in the accuracy of predictive distributions for births and the average fertility rate, for age-specific deaths and death probabilities, and for net migration and the associated rate. We assessed the goodness of fit of these predictive distributions from the UPE project against empirical data for the years 2003–2023 from Statistics Norway. The stock data are of very good quality. Data for birth and death counts have small errors; for net migration, the errors are larger. Appendix 1 gives a detailed assessment of the errors in the empirical data.

We reran PEP with point predictions and parameter settings for Norway. We computed $N = 1000$ simulations and used the same seed for the random number generator as in 2004. Therefore, our results are the same as those obtained in the UPE project. We stored the results for each calendar year and for single years of age. Empirical death probabilities $d(x,t) = D(x,t)/V(x,t)$ and net migration probabilities $n(x,t) = N(x,t)/V(x,t)$ were derived from actual counts, not approximate values as in (2) and (3). Average fertility rates $b(t)$ were computed as occurrence–exposure rates. We used age intervals 60–90 and 20–30 years for the analysis of mortality and net migration, respectively, for reasons explained in Appendix 1.

The populations exposed to childbearing risks $V^{cb}(t)$ (women aged 15–49 years) have a very different size compared to those involved in migration (ages 20–30, here) or the age groups used in the analysis of mortality (ages 60–90). Therefore, we used the scaled likelihood-ratio statistic (15), averaged across samples.

Fig. 4 shows the average goodness-of-fit statistic $\tilde{R}$ for deaths in ages 70, 75, 80, and 85 years, and births. For a given (x,t)-combination, $\tilde{R}$ has no intrinsic meaning: one can interpret it only by comparing two or more situations. A lower value of $\tilde{R}$ represents a relatively good fit, on average, between the predicted and empirical death probabilities and birth rates. Not surprisingly, Fig. 4 shows that the goodness of fit deteriorates when we look further into the future, and (for mortality) that it worsens with advancing age. Indeed, for age 90 (not shown in the graph) we found $\tilde{R}$ values increasing from 0.005 in 2003 to around 0.02 for the years 2020–2022. At age 85 years, the pattern is much more irregular than at ages 70, 75, and 80 years. This reflects stronger Poisson variability in the empirical death probabilities at age 85 years, caused by the small population size at this age.

Birth rates have $\tilde{R}$ values of the same magnitude as death probabilities around 75–80 years of age. Beyond 80 years old, death probabilities do worse than birth rates. But $\tilde{R}$ for fertility increases so fast that for forecast horizons much beyond 20 years, the fertility $\tilde{R}$ may very well be larger than the mortality $\tilde{R}$ for ages around 85 or even 90. This is caused by a generation effect: not only do the fertility rates have prediction errors but also women of childbearing age, because these women are increasingly born after the jump-off year of the forecast.

As noted above, net migration predictions are normally distributed. Hence, the predicted counts may be negative (or even zero after rounding to the accuracy used) for some samples, which means that the log likelihood (11) is undefined. Predictions with $N_{(i)}(x,t) \leq 0$ were omitted from the analysis. Under the normal model we use, net migration values ≤ 0 will always occur, albeit with very low probabilities in the applications we are interested in. Clearly, the model that was used for predicting migration is an approximation of reality at best.

Non-positive values occurred more often in ages above 30 years than below, and more often for predictions towards the end of the period from 2003–2022 than for the early years. In ages 20–30, the negative values were limited to less than 10% of all cases. Omitting cases with non-positive values implies that the accuracy of the migration forecast is overestimated.

Fig. 5 summarizes our results for net migration counts at ages between 20 and 30 years. Since the point predictions for $N(x,t)$ followed a smooth, slightly increasing path over time, the time pattern of the scaled $\tilde{R}$ values is dictated by the erratic actual numbers of net migration. Indeed, the time pattern of $\tilde{R}$ for age 25 years is very similar to that of the empirical numbers for ages 20–30 (see the dashed dark blue line). With a different choice of scale for the $\tilde{R}$ values, we can achieve a rather close correspondence with $\tilde{R}$ for age 20, or for age 30; see Fig. 6. The volatile character of empirical net migration is caused largely by immigration, with a strong increase in labor migration after the enlargement of the EU in 2004 and in 2007, a rapid fall after the global economic crisis in 2009, and the travel restrictions connected to the Covid-19 pandemic in 2020–2022.

The $\tilde{R}$ values in Figs. 5 and 6 are one order of magnitude larger than the $\tilde{R}$ values for deaths and births. The graphs illustrate an important point. The UPE forecasts of net migration are model-based, and the

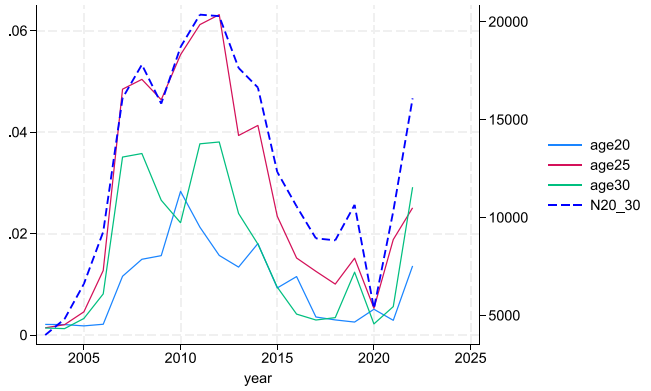


Fig. 5. Goodness-of-fit statistics $\tilde{R}$ for predictive distributions of net migration at ages 20, 25, and 30 years (solid colored lines, scale left). The dashed dark blue line shows empirical net migration counts for the age group 20–30 years (scale right). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

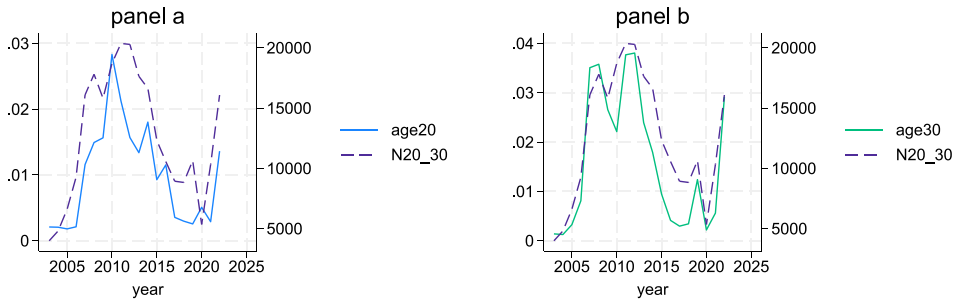


Fig. 6. Goodness-of-fit statistics $\tilde{R}$ for predictive distributions of net migration at ages 20 years (panel a) and 30 years (panel b). Note: different scales. The dashed dark blue line shows empirical net migration counts for the age group 20–30 years (scale right).

associated prediction intervals remain of limited width (although they become increasingly wider for longer forecast durations). Since actual migration shows strong non-stationary features, the goodness-of-fit statistic behaves erratically.

Thus, for this distance measure $\tilde{R}$ between predictive distributions for births and death counts and observed counts, and for this set of data from UPE predictions of fertility and mortality for Norway up to 20 years ahead, mortality (i.e. deaths) for ages around 75–80 years old was as difficult to predict as fertility (births). For higher ages, however, mortality forecasts are less accurate than those for fertility. Net migration flows are much more difficult to predict accurately than births or deaths.

7. Conclusion and discussion

We proposed a new measure (a score function) for evaluating the accuracy of predictive distributions of births, deaths, and net flows for international migration, when actual counts for these three types of events are available. Earlier attempts to assess the accuracy of probabilistic population forecasts focused on population size and age structure, not on the underlying three factors of population change. A novel approach infers predictions of births, deaths, and net migration from changes in predicted population stocks in carefully chosen age segments. In particular, the way we handled net migration is new.

The proposed score function is formulated in a common statistical framework for the three factors. It deals effectively with the way forecast parameters for fertility, mortality, and migration interact in the computation of population numbers. The score function is based on assumed Poisson distributions for the predicted and observed counts of the demographic events. It is closely related to likelihood-ratio-based scoring rules proposed earlier. We applied the measure to the results from a probabilistic population forecast for Norway. The base year of this forecast was 2003, and we assessed its accuracy for the years 2004–2023. Special attention was paid to the quality of the empirical data from Statistics Norway. We compared predictive distributions for age-specific deaths and death probabilities with empirical data for ages between 60 and 90 years, and likewise for the predictive distributions for births and general birth rates. We found that predictive distributions for deaths up to around age 75–80 are more accurate than such distributions for births. For ages beyond 80, the situation is reversed, and mortality distributions become less accurate than those for fertility. However, our findings suggest that for forecasts more than 15 years into the future, the fit of fertility predictions is worse than that of mortality even at ages around 90 years. Predicted counts for net migration in ages 20–30 years are much less accurate than those for births or deaths.

Our finding that fertility predictions up to 20 years ahead are more accurate than mortality predictions at age 80 years and over has not been reported before, to the best of our knowledge. Note that deterministic population forecasts computed in the second half of the previous century had surprisingly large errors in mortality; see Booth and Tickle (2008) and Oeppen and Vaupel (2002), and the references therein.

Our starting point was the assessment of forecasts of population flows and the related rates. The results can be extended to functions, or functionals, of the rates. Total fertility rate and life expectancy are two relevant examples.

Total fertility rate. In case the age distributions of the population in childbearing age, and those of age-specific fertility, do not change much over the forecast horizon, then the total fertility rate and the general fertility rate would be expected to behave similarly. This is realistic in the Norwegian case we used for illustration. Therefore, the assessment of forecast accuracy suggested for the general fertility rate would be expected to be realistic for the total fertility rate as well. Similar arguments hold for age-standardized mortality measures.

Life expectancy. Life expectancy is a highly nonlinear function of age-specific mortality rates, whose forecasting accuracy is detailed above. Shortcut methods are not available, but numerical methods can be used. They are computationally intensive, however, and they are not discussed here.

Data and code availability

All data and code necessary to reproduce the results gathered in the manuscript are available at https://github.com/NicoKeilman/GoFIJF.

CRediT authorship contribution statement

Juha Alho: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Nico Keilman:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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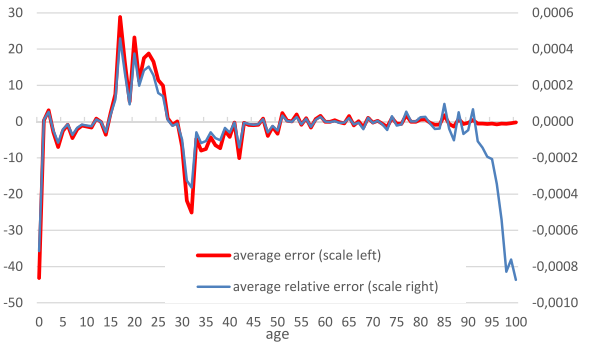


Fig. 7. Register errors: Averages for the period 2003–2022.

Appendix 1. Errors in population data

The Norwegian registration of births and deaths is considered to be of high quality (Vassenden, 2015). Errors occur primarily due to non-registration of out-migration. Krokedal et al. (2024) estimated that this concerned 20,000 known cases in the period from 2017–2021, or 0.44% of the population aged 15 years or older in 2022. Authorities eventually discover if an out-migrant failed to deregister. In such cases, the population registration system is corrected (Vassenden, 2015).

We compared stock and flow data for Norway for ages 0, 1, 2, …, 100 and the years 2003–2022. Counts $I(x,t)$ and $O(x,t)$ were combined into net counts $N(x,t) = I(x,t) - O(x,t)$. The starting point for the analysis is the bookkeeping Eq. (1), which in theory is a perfect equality. In empirical applications, errors in population registration systems and in population censuses lead to less-than-perfect equalities. We introduce an error term $E(x,t)$ (a register error) and write

$$V(x+1,t+1) = V(x,t) - D(x,t) + N(x,t) + E(x,t) \quad (16)$$

where $E(x,t)$ is the net effect of errors in all variables in (1). Fig. 7 summarizes our findings for ages $x \geq 0$.

Register errors are between -10 and $+10$ persons, with a few exceptions. They are larger for ages 17–25 years, and more negative for ages 30–35 and for age 0. Beyond age 50, the register errors are close to zero. In a relative sense (blue curve), the register errors are well below one per thousand, with strongly negative values for ages beyond 90 years, because increasingly fewer persons survive to a high age. The positive register errors for ages 17–25 appear to be due to timely registration by labor migrants and students. Negative errors for age group 30–35 years and for age 0 are compatible with delayed deregistration when leaving the country.

We define the approximation error for deaths as $D(x,t) - [V(x,t) - V(x+1,t+1)] = N(x,t) + E(x,t)$. Changing the roles of D and N, a corresponding measure for net migration is obtained.

Approximation errors for deaths are small for ages from 60 years onwards, both in size and relative to the actual death numbers; see Fig. 8. Negative errors for ages 61–68 are caused by net-outmigration for these ages, presumably in connection with retirement. For net migration,

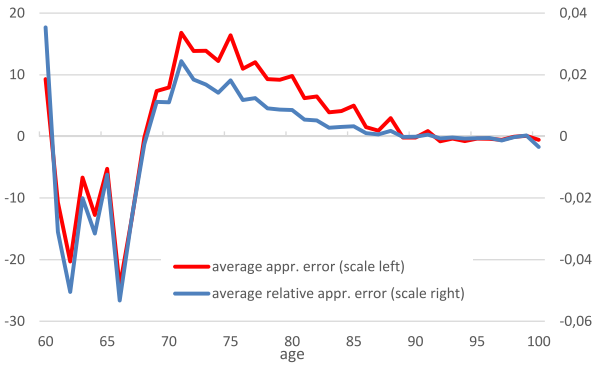


Fig. 8. Approximation errors for deaths. Age 60–100 years, averages for the period 2003–2022.

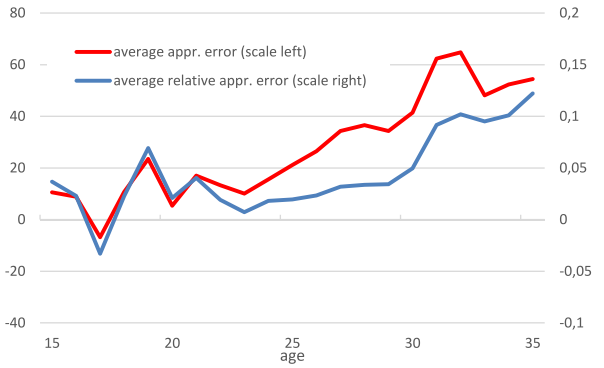


Fig. 9. Approximation errors for net migration. Age 15–35 years, averages for the period 2003–2022.

Fig. 9 shows average approximation errors for ages 15–35. They grow more or less regularly to around 60 persons on average for ages up to 35 years. This amounts to 12% of the actual value for that age. The errors increase rapidly for higher ages. We conclude that the approximation can be used for ages up to around 30 years, where the error is 5% or less.

Fig. 10 plots the actual live births (the blue curve) and approximate numbers (red curve). The approximation is very close.

To sum up: when following individuals along cohort lines, population growth in one-year age groups is close to the net effect of migration and mortality. The error is limited to between −30 and +30 persons, small as compared to the size of one-year age groups: 50,000–60,000 for ages up to 60 years. Beyond age 60, the register error is close to zero. Thus, the number of deaths is well approximated by population change for ages 60 and higher. For net migration, population change can be used for ages 20–30 years, where the error is less than 5%. Approximating live births by the population aged 0 at the end of the year implies errors of less than 1%.

Appendix 2. Poisson nature of in-migration

Consider $I(x,t)$. It is a sum of directed out-migrations from all other countries: $I(x,t) = \sum_c O_c^*(x,t)$, where

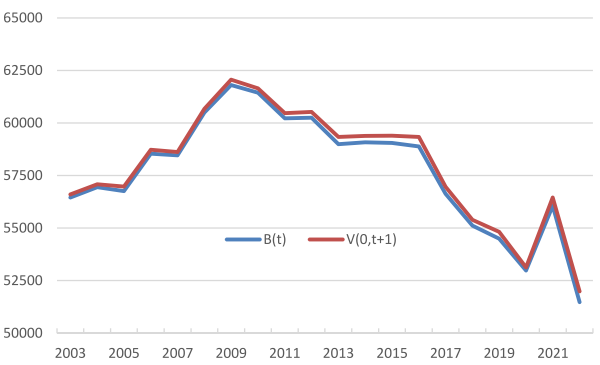


Fig. 10. Live births from 2003–2022: actual numbers $B(t)$ and approximated numbers $V(0,t+1)$.

$O_c^*(x,t)$ is the number of out-migrants from country c to our country of interest, such that $O_c^*(x,t) \sim Bin(V_c(x,t), o_c^*(x,t))$ are independent. Suppose the probabilities for country c are of the form $o_c^*(x,t) = \kappa_c(x,t)V(x,t)$, where $\kappa_c(x,t) = o_c(x,t)/(V(x,t) + \sum_{j\neq c} V_j(x,t))$. This supports the Poisson approximation $I(x,t) \sim Po(E[I(x,t)])$, where $E[I(x,t)] = \sum_c o_c^*(x,t)V_c(x,t) = V(x,t)\iota(x,t)$ and where $\iota(x,t) = \sum_c \kappa_c(x,t)V_c(x,t)$.

Appendix 3. Notes on poisson likelihood

Suppose count Y has a Poisson distribution, $Y \sim Po(\lambda)$. When $Y = y$ is observed, the likelihood is $L(\lambda) = e^{-\lambda}\lambda^y/y!, y = 0,1,2,\ldots$ The log likelihood is, up to an additive constant, $\ell(\lambda) = -\lambda + y\log(\lambda)$. The derivative is $\ell'(\lambda) = -1 + y/\lambda$. Setting this to zero yields the MLE $\hat{\lambda} = y$. The second derivative is $\ell''(\lambda) = -y/\lambda^2$.

Consider the hypothesis $H_0 : \lambda = \lambda_0$. The likelihood-ratio test statistic is $R = 2(\ell(y) - \ell(\lambda_0))$. The second-order Taylor series approximation, at $\lambda = y$, to the latter term is $\ell(\lambda_0) = \ell(y) + \ell'(y)(\lambda_0 - y) + \frac{1}{2}\ell''(\tilde{\lambda})(\lambda_0 - y)^2$, where $\ell'(y) = 0$, and $\tilde{\lambda}$ is some number between y and λ_0. Then, the test statistic can be written as $R = C(Z + \kappa)^2$, where $C = y\lambda/\tilde{\lambda}^2, Z = (y - \lambda)/\sqrt{\lambda}$, and $\kappa = (\lambda - \lambda_0)/\sqrt{\lambda}$.

The normal approximation to the Poisson distribution means that $Z \sim N(0,1)$ asymptotically. Since $E[Y/\lambda] = 1$ and $Var(Y/\lambda) = 1/\lambda \downarrow 0$ as $\lambda \to +\infty$, also $Y/\lambda \to 1$ in probability. Then also $C \to 1$ in probability. The conclusion is that for large λ, $R \approx (Z + \kappa)^2$ has an approximately non-central χ^2 distribution with one degree of freedom. This reflects both the Poisson variation in the counts of the events and the bias of the forecast via the non-centrality parameter κ^2.

This characterization pertains to a single pair (x,t) and a single simulation round from the predictive distribution. It is offered here to provide some intuition about the meaning of the proposed measures. In fact, the measures for different simulation rounds are closely related if they pertain to a single observable count, so no similar claims about the stochastic characteristics of the average (over simulation rounds) can be made.

On the other hand, when such measures are added (over age x, or time t, or both), then one can expect that for a single simulation round, the sum also has an approximately non-central χ^2 distribution with degrees of freedom equal to the number of summands and a non-centrality parameter equal to the sum of non-centrality parameters of summands. But again, after averaging, this simple characterization can no longer be claimed.

Below, we repeat part of the explanation of Section 5.2 in order to clarify the argument.

When data from different settings are compared, the role of the non-centrality parameter becomes important. To simplify, let us omit the dependency on age and time (x, t), so in the case of mortality, the role of Y is taken by the death count D, with the expectation $\lambda = \pi V$, so the MLE is $\hat{\pi} = D/V$. If V is very large (or small), then D would typically be large (small) as well. For the purpose of comparing the accuracies of different mortality forecasts, we may consider adjusted data, substituting $V := cV$ and $D := cD$ for some $c > 0$. This would leave the MLE unchanged, $\hat{\pi} = D/V$, as before, but the numerical value of R would be replaced by cR.

In the adjusted scale, we would interpret $\lambda_0 = d_{(i)}cV$, so $\kappa = \sqrt{cV}(\pi - d_{(i)})/\sqrt{\pi}$, for the ith simulation round. Consider two possible values of c. (1) Taking $c = 1/V$, we would have $\kappa^2 = (\pi - d_{(i)})^2/\pi$, which measures bias in what one might call chi-squared scale, independently of the original size of the population. (2) Set $c = 1/D \approx 1/\pi V$, where a large population approximation is intended. In this case, we would have $\kappa^2 \approx (1 - d_{(i)}/\pi)^2$, which measures the squared relative error, also independently of the size of the original population.

In the process of adjusting, the size of the term representing Poisson variation also changes. This may or may not be what is desired. It all depends on the Assessor's interests.

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